MTH212M End Semester Examination
Part A (February 25, 2026)

5:00 pm - 6:15 pm (+25 minutes for DAP students)
Maximum marks for Part A: 40
Minimum marks for Part A: 0

Instructions:

1. DO NOT attach any additional sheet to this page.
2. Write your answers in the designated place. If your answer is not legible, you shall not get credit.
3. If necessary, you may write your answer using a simple fraction.
4. All sub-questions carry **negative marks (-1)** for an incorrect answer and **0 marks** for no attempt.
5. Statements written outside designated spaces shall not be evaluated.
6. Please handle this sheet carefully. No replacement shall be provided.

Name:

Roll No.:

Question 1. (2 + 1 marks) Let i be an aperiodic, positive recurrent state in a Markov Chain with the mean recurrence time 2. Then, $\boxed{\lim_{n\to\infty} p_{ii}^n = \qquad\qquad}$ and $\boxed{\lim_{n\to\infty} \frac{1}{n}\sum_{m=1}^{n} p_{ii}^m = \qquad\qquad}$

Question 2. (1 + 2 marks) Let i be a positive recurrent state in a Markov Chain with the mean recurrence time 4. Suppose period of i is 2. Then, $\boxed{\lim_{n\to\infty} p_{ii}^n = \qquad\qquad}$ and $\boxed{\lim_{n\to\infty} \frac{1}{n}\sum_{m=1}^{n} p_{ii}^m = \qquad\qquad}$

Question 3. (1 + 1 marks) Let i be a null recurrent state in a Markov Chain with period 3. Then, $\boxed{\lim_{n\to\infty} p_{ii}^{3n} = \qquad\qquad}$ and $\boxed{\lim_{n\to\infty} p_{ii}^{3n+1} = \qquad\qquad}$

Question 4. (1 mark) Suppose i and j be two states in a Markov Chain such that i is recurrent and $i \leftrightarrow j$. Then, $\boxed{f_{ij} = \qquad\qquad}$

Question 5. (1 mark) Suppose that a recurrent Markov Chain with the state space $S = \{0, 1, 2, \dots\}$ is known to be irreducible. If $\sum_{n=1}^{\infty} n f_{11}^n = \infty$, then the number of null recurrent classes in this Markov Chain is $\boxed{\qquad\qquad}$

Question 6. $\{X_n\}_{n=0}^{\infty}$ be a Markov Chain on the state space $S = \{0, 1, 2, 3, 4, 5\}$ and with the transition probability matrix

$$P = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} & 0 & 0 & 0 & 0 \\ \frac{2}{3} & \frac{1}{3} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{3}{4} & \frac{1}{4} & 0 & 0 \\ \frac{1}{4} & 0 & \frac{1}{4} & 0 & 0 & \frac{1}{2} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{3} & 0 \end{pmatrix}$$

(a) (2 marks) The number of communication classes is $\boxed{\qquad\qquad}$

(b) (1 mark) The number of communication classes, which are closed, is $\boxed{\qquad\qquad}$

(c) (2 marks) The number of positive recurrent classes is $\boxed{\qquad\qquad}$

Question 7. Underline 'Yes' or 'No' as appropriate in the following sub-questions.

(a) (2 marks) In an aperiodic Markov Chain with finite state space S and transition probability matrix $(p_{ij})_{i,j\in S}$, it is known that there is a transient state. Consider the statement: 'There exists a proper subset $C \subsetneq S$ such that $(p_{ij})_{i,j\in C}$ is stochastic.' Is the statement true? $\boxed{\text{Yes / No}}$

(b) (2 marks) Consider any Markov Chain $\{X_n\}_{n\geq 0}$ with a countably infinite state space and with exactly one absorbing state and at least one transient state. Let i and j denote the absorbing and a transient state, respectively. Consider the statement: 'There exists examples of such Markov Chains such that $\mathbb{P}(\{X_n\}_{n\geq 1}$ is absorbed into $i \mid X_0 = j) < 1$.' Is the statement true? $\boxed{\text{Yes / No}}$

Question 8. In this problem, we consider red and green balls. Balls are identical except for colour and balls of the same colour are identical. Suppose that an urn contains one red ball and two green balls. Consider the following random experiment: a ball is drawn from the urn at random and is replaced with a new ball of the other colour, i.e. a drawn red ball is replaced by a new green ball and vice versa. At the time $n = 0$, we do not perform this experiment and instead look at the initial state as mentioned earlier. At the time points $n = 1, 2, \ldots$, the random experiment is performed once. Denote by X_n the number of red balls in the urn at the time $n = 0, 1, 2, \ldots$ and consider the Markov Chain $\{X_n : n = 0, 1, 2, \ldots\}$.

(a) (1 mark) The state space is $\boxed{}$

(b) (2.5 marks) The transition probability matrix is

$$\begin{pmatrix} \\ \\ \\ \end{pmatrix}$$

(c) (1 mark) Period of state 0 is $\boxed{}$

(d) (1 mark) $\mathbb{P}(X_{n+2} = 2 \mid X_n = 1) =$ $\boxed{}$

(e) (1 mark) Is the Markov Chain irreducible? Underline the correct answer. $\boxed{\text{Yes / No}}$

(f) (2.5 marks) Write down a stationary distribution for this Markov Chain as a row vector $\boxed{}$

(g) (1 mark) Is the above stationary distribution unique? Underline the correct answer. $\boxed{\text{Yes / No}}$

Please turn over

Question 9. Consider a Markov chain $\{X_n\}_{n=0}^{\infty}$ with state space $S = \{0, 1, 2, 3, 4\}$ and the transition probability matrix

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ \frac{1}{4} & 0 & \frac{3}{4} & 0 & 0 \\ 0 & \frac{1}{4} & 0 & \frac{3}{4} & 0 \\ 0 & 0 & \frac{1}{4} & 0 & \frac{3}{4} \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

(a) (1 mark) Period of state 2 is

(b) (1 mark) Period of state 0 is

(c) (4 marks) $\mathbb{P}(\{X_n\}_{n\geq 1} \text{ is absorbed into } \{4\} \mid X_0 = 3) =$

(d) (1 mark) The number of positive recurrent states is

(e) (1 mark) The number of transient classes is

(f) (2 marks) Write down a stationary distribution for this Markov Chain as a row vector

(g) (1 mark) Is the above stationary distribution unique? Underline the correct answer. Yes / No